

CSE 4412: Introduction to Quantum Computing - Week 2 Notes

Postulate 2: measurements

This is how we can extract information from a quantum system. We can measure a quantum state $|\psi\rangle$ by applying a measurement operator M to it. The measurement operator is a set of projectors $\{P_i\}$ that correspond to the possible outcomes of the measurement. When we apply the measurement operator to the state, we get a new state that is one of the possible outcomes, and the probability of getting that outcome is given by the Born rule: $P(i) = \langle\psi| P_i |\psi\rangle$.

where $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$. where $\mathbb{P}_i$ is the probability of measuring the state $|\psi\rangle$ in the state $|i\rangle$, and $\langle\psi| P_i |\psi\rangle$ is the inner product of the state $|\psi\rangle$ with the projector P_i applied to it. Given that they are bases we can rewrite $|0\rangle$ and $|1\rangle$ in terms of $|+\rangle$ and $|-\rangle$ as follows:

$$|0\rangle = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle), \text{ and } |1\rangle = \frac{1}{\sqrt{2}}(|+\rangle - |-\rangle)$$

Postulate 4: How to model two or more states

We can use tensor product to model two or more quantum states. If we have two quantum states $|\psi_1\rangle$ and $|\psi_2\rangle$, we can combine them into a single state using the tensor product:

$$|\psi_1\rangle \otimes |\psi_2\rangle$$